

Multidisease Deep Learning Neural Network for the Diagnosis of Corneal Diseases



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• **ABSTRACT:** • **PURPOSE:** To report a multidisease deep learning diagnostic network (MDDN) of common corneal diseases: dry eye syndrome (DES), Fuchs endothelial dystrophy (FED), and keratoconus (KCN) using anterior segment optical coherence tomography (AS-OCT) images.

• **STUDY DESIGN:** Development of a deep learning neural network diagnosis algorithm.

• **METHODS:** A total of 158,220 AS-OCT images from 879 eyes of 478 subjects were used to develop and validate a classification deep network. After a quality check, the network was trained and validated using 134,460 images. We tested the network using a test set of consecutive patients involving 23,760 AS-OCT images of 132 eyes of 69 patients. The area under receiver operating characteristic curve (AUROC), area under precision-recall curve (AUPRC), and F1 score and 95% confidence intervals (CIs) were computed.

• **RESULTS:** The MDDN achieved eye-level AUROCs >0.99 (95% CI: 0.90, 1.0), AUPRCs > 0.96 (95% CI: 0.90, 1.0), and F1 scores > 0.90 (95% CI: 0.81, 1.0) for DES, FED, and KCN, respectively.

• **CONCLUSIONS:** MDDN is a novel diagnostic tool for corneal diseases that can be used to automatically diagnose KCN, FED, and DES using only AS-OCT images. (Am J Ophthalmol 2021;226: 252–261. © 2021 Published by Elsevier Inc.)

Corneal disease is a major cause of visual impairment and ocular discomfort that imposes a very significant burden on patients' quality of life as well as the public health system.^{1–3} Dry eye syndrome (DES), Fuchs endothelial dystrophy (FED), and keratoconus (KCN) are among the most common corneal diseases. DES has a massive socioeconomic burden, costing US \$700,000 per mil-

lion patients.⁴ Even though there are several diagnostic tests available, no gold standard test has been described in the diagnosis of DES.⁵ FED and pseudophakic bullous keratopathy are the leading indications for corneal transplantation in the United States.⁶ The latter group usually consists of patients with undiagnosed FED, where early diagnosis is critical to determining the management and surgical approach option. FED is responsible for 36% of performed corneal transplants in the US, whereas the prevalence of the disease ranges from 3.8% to 11% of patients over the age of 40, depending on the geographic region.^{7,8} KCN is the third most common indication for corneal transplantation with an overall prevalence rate of 54.5 per 100,000 in the United States.^{9,10} It is noteworthy that the second most common cause of corneal transplantation is repeat corneal transplantation secondary to corneal graft failure and that sheds the light on the impact of FED and KCN on the public health system.⁹

Screening and subsequently the early diagnosis and treatment of corneal diseases are critical to decrease the burden of the disease on patient quality of life and the public health system.¹¹ A comprehensive screening of the cornea is costly and time-consuming, requiring multiple diagnostic devices such as slit-lamp biomicroscopy, Scheimpflug tomography, specular microscopy, and pachymetry. The increase in disease prevalence and the scarcity of resources is a gap that autonomous diagnosis using artificial intelligence (AI) can potentially bridge.

Since its first introductory use for the retina, optical coherence tomography has found further purpose as an essential tool in the diagnosis of anterior segment diseases. Anterior segment optical coherence tomography (AS-OCT) is capable of noninvasively providing a cross-sectional high-resolution image of all the corneal microlayers and detecting microscopic features of corneal diseases such as FED, KCN, and DES.^{12–14} AS-OCT poses as a device that have the potential, if coupled with autonomous diagnostics algorithms, to replace the whole battery of tests used to diagnose corneal diseases in ophthalmology clinics as well as allowing using less specialized health care personnel and facilities to reduce the burden of the disease.

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Artificial intelligence (AI) is a field of study that aims to create machines that operate with human-level intelligence.¹⁵ A subfield of interest in AI is machine learning which aims to create a set of algorithms able to learn hidden patterns existing in data. Recent advances in AI have led to creating a new class of algorithms, based on artificial neural networks (ANNs), called deep learning.¹⁶ The power of deep learning algorithms comes from their ability to learn a hierarchy of features with different levels of abstraction from the existing data, which makes them exhibit large discriminative power. In ophthalmology, deep learning has been used for screening diseases such as diabetic retinopathy, glaucoma, age-related macular degeneration, and retinopathy of prematurity using color fundus images or OCT images.^{17,18} Recently, a deep learning algorithm, IDx-DR (IDx Technologies; Coralville, Iowa, USA), using a retinal camera has been approved by the Food and Drug Administration (FDA) for the screening of referral-warranted diabetic retinopathy.¹⁹ Existing AI studies on corneal pathologies are lagging behind with only reports of single disease models for diagnosis of keratoconus and graft detachment after Descemet membrane endothelial keratoplasty (DMEK).²⁰⁻²³ As of this writing, there has been no multidisease deep learning algorithm for the automated detection of the common corneal diseases, a critical step in exploiting the power of AI to allow automated screening for corneal diseases and reducing their burden on patient quality of life as well as the public health system.

In this study, we developed deep learning models to automatically diagnose and screen for DES, FED, and KCN from healthy corneas using AS-OCT images. First, we have created a data set of corneal AS-OCT images by prospectively recruiting patients with corneal diseases presented to Bascom Palmer Eye Institute over a 40-month period. Second, we used transfer learning to develop and validate a multidisease deep learning model for screening DES, FED, and KCN. Third, we tested the model in the setting of an additional independent clinical study. Fourth, we compared the performance of the multidisease deep learning model to 3 single-disease models for each disease. Our findings elaborate on the efficacy of our deep learning algorithm in the autonomous diagnosis of corneal diseases using AS-OCT images.

METHODS

• **STUDY DESIGN AND PARTICIPANTS:** This study was approved by the University of Miami Institutional Review Board. The study design complied with the Health Insurance Portability and Accountability Act (HIPAA), and the research was conducted under the Declaration of Helsinki. All subjects provided written informed consent before participation. A total of 158,220 AS-OCT images (45,900

healthy; 16,740 FED; 64,800 keratoconus; 30,780 DES) of 879 eyes of 478 participants, prospectively recruited from June 2016 to September 2019 at the Bascom Palmer Eye Institute, were used to develop, validate, and test a multidisease deep learning diagnostic network (MDDN) of common corneal diseases, namely, DES, FED, and KCN using AS-OCT images. All participants underwent a thorough ophthalmologic examination by one of 6 fellowship-trained American Board of Ophthalmology–certified cornea specialists, masked to the automated diagnosis given by the algorithm, to assign each cornea to one of the aforementioned diagnoses. The diagnosis was made at the eye level and all images captured from the eye were labeled with the clinical diagnosis. A healthy patient was defined as having no ICD-10 corneal diagnoses, a patient with DES was defined as having an ICD-10 diagnosis of DES (H04.123), FED was defined as having an ICD-10 diagnosis of FED (H18.51), and KCN was defined as having an ICD-10 diagnosis of KCN (H18.603). Patients with other ICD-10 corneal pathologies were excluded from all groups. The data that support the findings of this study are available on request from the corresponding author (M.A.S.). The data are not publicly available because of intellectual property considerations.

• **IMAGE ACQUISITION:** AS-OCT (Envisu R2210, LEICA, Buffalo Grove, Illinois, USA) was performed on each participant. Scans were obtained using a 6-mm radial scan pattern with 36 cross-sectional images repeated 5 times (ie, 180 in total). Capturing 180 frames and not 10 frames per eye allowed creating a larger data set to train our neural network. Larger training set likely improves the robustness of the neural network. It is noteworthy that the OCT device is capable of capturing these 180 frames in few seconds. For the testing, there is likely less benefit in capturing 180 frames per eye than in training. Future study that identifies the optimal number of frames per eye is warranted. Scans were taken for the central cornea, using the apex of the cornea as a focus point to ensure centralization of the scan. The existence of central saturation artifact ensures the scan is centered.^{24,25} Basic demographic information, dates, times, and locations of patients during visits and scans, and other clinical variables were extracted from the University of Miami electronic medical record (EMR) system. The patients were organized by clinical category—healthy, DES, FED, and KCN. The data were stripped of all protected health identifiers. For the training, validation, and testing procedures, all images were anonymized.

• **NETWORKS:** We used a pre-trained VGG19²⁶ network for the diagnosis of corneal diseases vs healthy controls. The VGG19 network consists of 14 convolutional layers for feature extraction, five MaxPooling layers for feature reduction (ie, maximum pooling), two fully connected layers of size 4096, and a classification layer with a Softmax activation function and 1000 categories. The detailed ar-

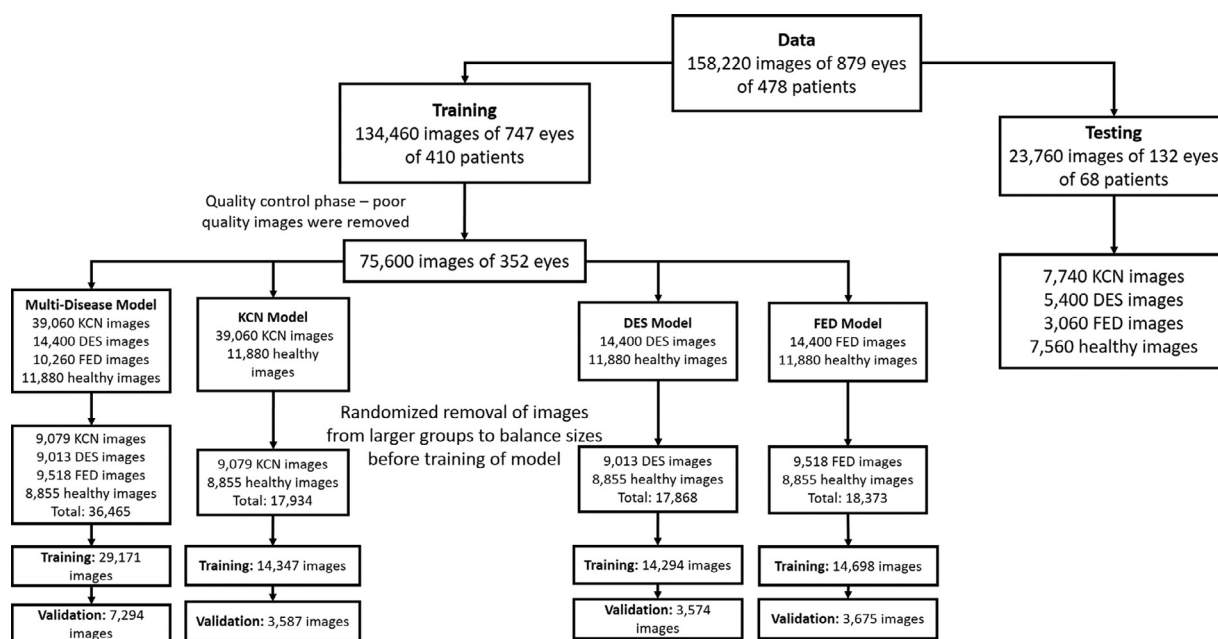


FIGURE 1. Distribution of the corneal optical coherence tomography images of the training, validation, and testing data sets for the Multidisease Diagnostic Model and the Single-Disease Diagnostic Models.

chitecture of the VGG19 network is shown in Figure 2 in the supplementary material. Some diseases can co-occur together, for example, FED and DES, which require a multilabel classification network to produce multiple outputs. We implemented the multilabel network using two approaches. First, we used multiple outputs and replaced the Softmax layer with sigmoid layer. Second, we split the eyes with FED into two categories, FED and FED & Dry, which enabled us to use Softmax layer. In addition, we developed a network only to provide the primary diagnosis, where we replaced the classification layer by a new classification layer with a number of categories that fit our classification task. We fine-tuned the classification network using our training data, which is known as transfer learning with fine-tuning. For the MDDN, we used four categories in the classification layer (ie, DES, FED, Healthy, and KCN). To further evaluate our MDDN diagnostic performance, we trained another MDDN for repeatability using a different random initialization (labelled MDDN 2 below). Also, we developed single-disease deep learning diagnostic networks (SDDN) to discriminate only one disease from the healthy controls by using a classification layer with two categories (ie, Healthy vs Disease). All models were trained, validated, and tested using the same data sets. For convenience, we report the performance of the MDDNs in the main article and the performance of the SDDNs in the supplementary material.

• **TRAINING AND TESTING:** Figure 1 provides the details of the training and testing sets. We used 134,460 images of 747 eyes of 410 patients in the training and testing. Image quality check was executed by trained operators (T.E., E.O.,

C.C., and M.T.) by removing images that contained any of the following: decentralization, blinking, missing parts of the endothelium or epithelial layers of the cornea, or low signal-to-noise ratio. After curating the data set for quality, 75,600 images from 484 eyes were selected for the training. A balanced set of images (ie, with a close number of images for each category) was selected from each category (36,465 images from 352 eyes [9,079 KCN, 9,013 DES, 9,518 FED, and 8,855 healthy]; Figure 1). The selected images were randomly split into 80% for training and 20% for validation. The developed models were trained using single AS-OCT images labeled by the diagnosis of the eye. The output of the model is a diagnosis for each AS-OCT image.

Subsequent to training and validation, the developed models were tested in the setting of an independent clinical study on different consecutive patients attending the cornea clinic of Bascom Palmer Eye Institute from September 2019 to October 2019. We obtained 23,760 AS-OCT images (7,560 healthy, 3,060 FED, 7,740 keratoconus, and 5,400 DES) prospectively from 132 eyes of 68 patients captured using the same OCT machine. Figure 1 summarizes the data sets used with the developed networks. During testing, the trained models were tested to give the diagnosis on individual images (ie, image-level). The eye diagnosis is, then, obtained using the majority of the OCT scan diagnoses obtained from each eye. We used a diagnostic optimal cutoff value of 50% frames plus one (ie, majority) for all the categories—except for KCN; because of its focal nature, we used an optimal cutoff value of 16% frames minimum. The optimal cutoff value was determined in the development stage of the algorithm.

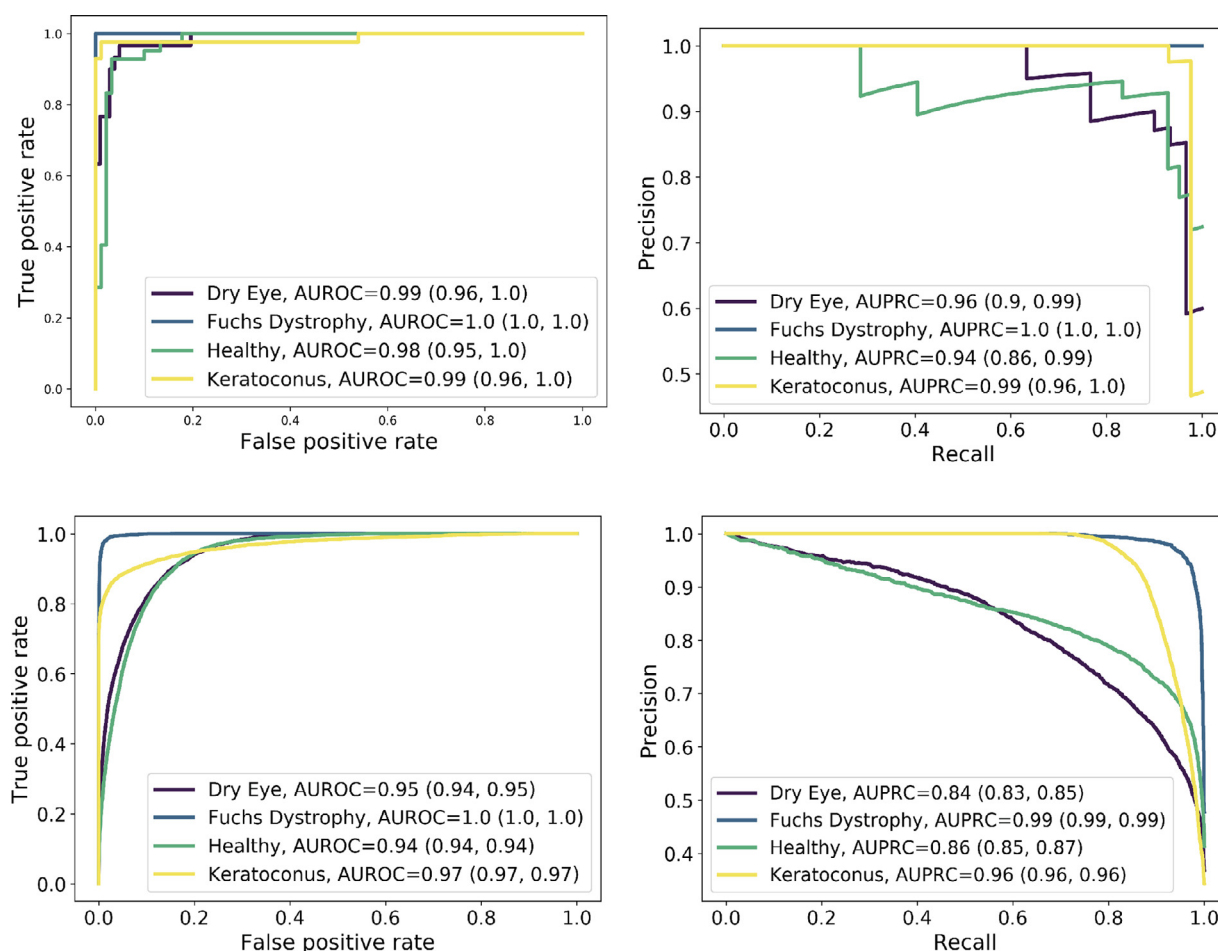


FIGURE 2. The eye-level receiver operating characteristic (ROC) and precision-recall (PR) curves for each category (top) and the image-level ROC and PR curves for each category (bottom). The area under the ROC curve (AUROC) and the area under the PR curve (AUPRC) for each category are displayed in the legend of the plots.

- **VISUALIZING THE MODELS:** To visualize the image pixels that were important for the network to get the diagnosis of the input AS-OCT image, we computed a heatmap using the weights of the last layers of the network for each input AS-OCT image and overlaid the heatmap on the image. Also, we performed an occlusion test by sequentially replacing patches in an image with random values and computing the probability of detecting the disease by our network.²⁷

- **IMPLEMENTATION AND STATISTICAL ANALYSIS:** All networks were developed using MATLAB 2019b (MathWorks, Natick, Massachusetts, USA). The models were trained by optimizing a cross entropy loss function using stochastic gradient descent (SGD) optimizer with a learning rate of 0.0001 and a squared-norm weight regularization rate of 0.0001. All models were trained for 10 epochs (ie, loops on data). Statistical analyses were performed using SPSS software, version 25.0 (Armonk, New York, USA) and MATLAB 2019b (MathWorks). We plotted the receiver-operating characteristic (ROC) and the

precision-recall (PR) curves, and computed the area under the ROC curve (AUROC) and the area under the PR curve (AUPRC) with 95% confidence intervals for each class on both image level and eye level. Also, we reported the confusion matrices for both the image level and eye level.

RESULTS

- **PARTICIPANTS:** Our study included 879 eyes of 478 participants who were prospectively and consecutively recruited from June 2016 to October 2019 at Bascom Palmer Eye Institute. The demographics and the breakdown into each category are summarized in the supplementary material.

- **TRAINING AND VALIDATION SETS:** For the training of the MDDN, there was a continuous decrease in the loss function (ie, categorical cross entropy) of the validation

TABLE 1. The Eye-Level and Image-Level Prediction Metrics With 95% Confidence Intervals for DES, FED, HL, and KCN Using the Multidisease Diagnostic Model

Model	Metric	DES		FED		HL		KCN	
		Eye	Image	Eye	Image	Eye	Image	Eye	Image
MDDN 1	F1 score	0.90	0.76	1.0	0.96	0.93	0.77	0.98	0.89
	Lower	0.81	0.75	1.0	0.95	0.86	0.76	0.94	0.89
	Upper	0.97	0.77	1.0	0.96	0.98	0.77	1.0	0.90
	AUROC	0.99	0.95	1.0	1.0	0.98	0.94	0.99	0.97
	Lower	0.97	0.95	1.0	1.0	0.95	0.94	0.96	0.97
	Upper	1.0	0.95	1.0	1.0	1.0	0.94	1.0	0.97
	AUPRC	0.96	0.84	1.0	0.99	0.94	0.86	1.0	0.96
	Lower	0.90	0.83	1.0	0.99	0.86	0.85	0.96	0.96
	Upper	0.99	0.85	1.0	0.99	0.99	0.87	0.99	0.96
MDDN 2	F1 score	0.86	0.76	1.0	0.97	0.88	0.75	0.97	0.88
	Lower	0.76	0.75	1.0	0.97	0.79	0.74	0.91	0.88
	Upper	0.94	0.77	1.0	0.97	0.95	0.76	1.0	0.89
	AUROC	0.98	0.95	1.0	1.0	0.97	0.94	0.99	0.97
	Lower	0.95	0.95	1.0	1.0	0.94	0.93	0.97	0.97
	Upper	1.0	0.95	1.0	1.0	0.99	0.94	1.0	0.97
	AUPRC	0.94	0.84	1.0	1.0	0.92	0.85	0.99	0.96
	Lower	0.88	0.84	1.0	1.0	0.83	0.84	0.96	0.95
	Upper	0.99	0.85	1.0	1.0	0.98	0.86	1.0	0.96

AUPRC = area under the precision-recall curve, AUROC = area under the receiver operating characteristic curve, DES = dry eye syndrome, FED = Fuchs endothelial dystrophy, HL = healthy, KCN = keratoconus.

data set, which ensured that the model was learning. As the model was learning, the classification accuracy of the validation data set was increasing (see the training loss and accuracy figure in the supplementary material).

• **TEST SET:** The eye-level and image-level ROC and PR curves were plotted for the MDDN 1 as shown in Figure 2. Also, the results of the MDDN 1 and MDDN 2 on the test set are summarized in Table 1. On the eye level, the MDDN 1 achieved AUROCs >0.99 (95% CI: 0.96, 1.0), AUPRCs >0.96 (95% CI: 0.90, 1.0), and F1 scores >0.90 (95% CI: 0.81, 1.0). On the image level, the MDDN 1 achieved AUROCs >0.95 (95% CI: 0.95, 1.0), AUPRCs >0.84 (95% CI: 0.83, 0.99), and F1 scores >0.76 (95% CI: 0.75, 0.96).

On the eye level, the MDDN 2 achieved AUROCs >0.98 (95% CI: 0.95, 1.0), AUPRCs >0.95 (95% CI: 0.88, 1.0), and F1 scores >0.86 (95% CI: 0.76, 1.0). On the image level, the MDDN 2 achieved AUROCs >0.95 (95% CI: 0.95, 1.0), AUPRCs >0.85 (95% CI: 0.84, 1.0), and F1 scores >0.76 (95% CI: 0.75, 0.97). The multilabel models achieved inferior results to the MDDN that provide the primary diagnosis. Labeling eyes with both FED and dry eye introduced imbalance in the data and caused confusion between FED and other categories. We put the test results of the multilabel models in the supplementary material.

The intraclass correlation analysis results between MDDN 1 and MDDN 2 are summarized in Table 2. The results show excellent reliability between the 2 models with

TABLE 2. Intraclass Correlation Analysis Between the 2 Trained Models With 95% Confidence Interval

		Intraclass Correlation	95% Confidence Interval	
			Lower Bound	Upper Bound
Single measures	Eye	0.89	0.85	0.92
	Image	0.85	0.85	0.86
Average measures	Eye	0.94	0.92	0.96
	Image	0.92	0.92	0.92

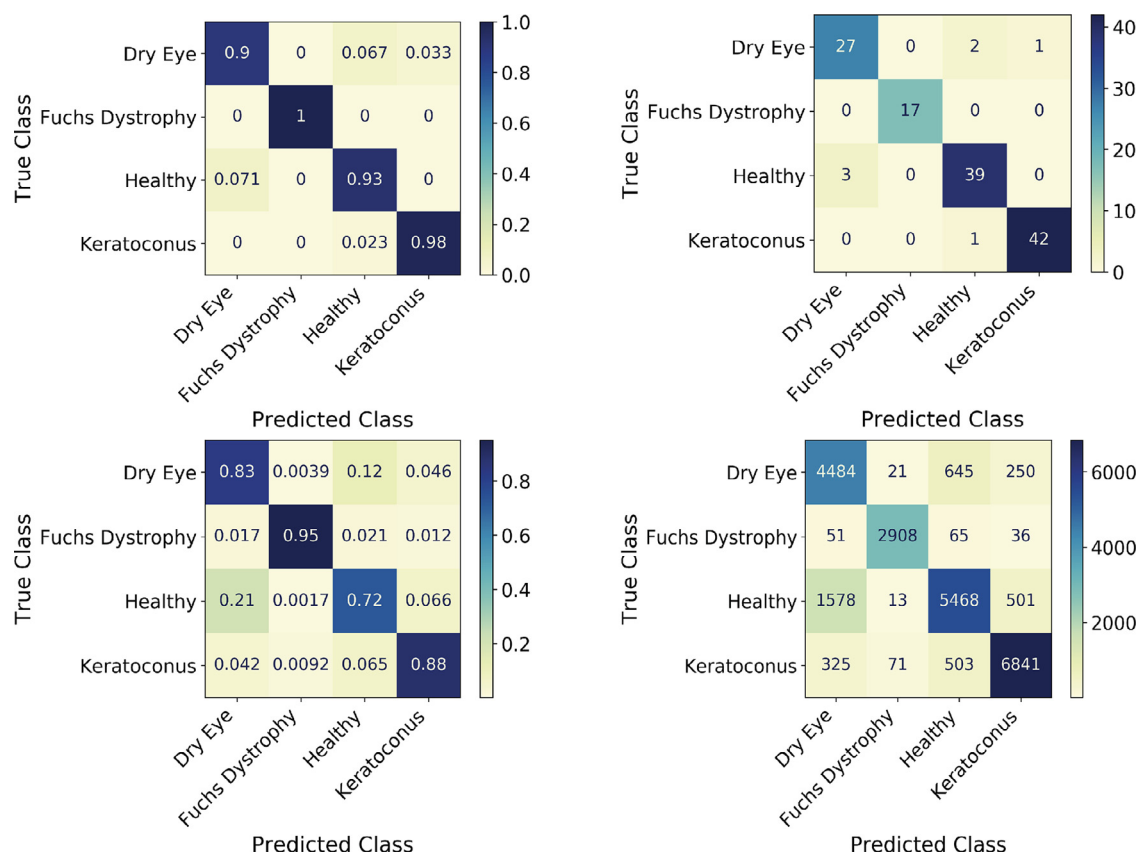


FIGURE 3. The eye-level normalized and raw confusion matrices (top) and the image-level normalized and raw confusion matrices (bottom).

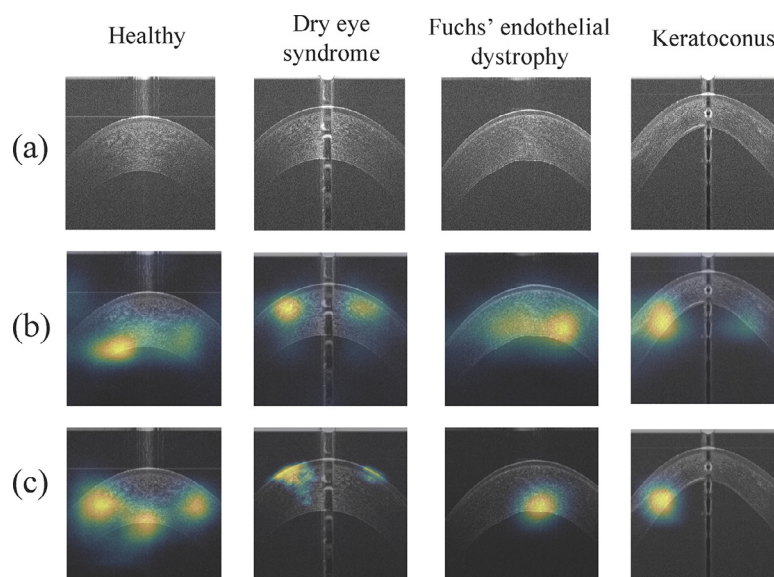


FIGURE 4. (A) Examples of the corneal optical coherence tomography (OCT) images for each category, (B) visualizations of the weighted averaged activations learned by the Multidisease Diagnostic Model (MDDM) for each corneal OCT image, and (C) visualizations of the results of the occlusion test using the MDDM.

intraclass correlations of 0.89 on eye level and 0.85 on image level.

The normalized and raw confusion matrices are shown in Figure 3 for both eye level and image level. In Figure 3, the eye-level confusion matrices show the model predictions by clinical diagnoses for 132 eyes. In 125 eyes, the model prediction agreed with the clinical diagnosis. The MDDN made 7 errors in diagnosis at the eye level. Five errors were confusion between dry eye and healthy classes. The sixth error was confusion between dry eye and keratoconus classes. The seventh error was confusion between keratoconus and healthy classes. Of note, the other eye of this patient was correctly classified as KCN.

Visualizations of the weighted activation maps and occlusion test of the MDDN are shown in Figure 4 for some OCT images. From the visualizations, the MDDN highlighted the characterizing regions for each disease, for example, the posterior surface of the cornea was highlighted for FED.

DISCUSSION

This study represents the first effort to develop and validate the performance of deep learning multidisease algorithm in the autonomous diagnosis of common corneal diseases with significant morbidity and economical burden. The total cost of DES on the health care system in the United States was estimated at US \$3.84 billion, with increasing direct costs with disease progression.²⁸ According to a 2016 statistical study from the Eye Bank Association of America (EBAA), FED was the most common indication of the corneal transplantation and accounted for 93% of the overall corneal transplants performed to correct endothelial cell failure.²⁹ According to the same study, KCN was the third most common indication for corneal transplantation.²⁹ KCN, a disease with an estimated prevalence of 1 in 375 individuals, led to more than 20% of patients requiring corneal transplantation to regain visual acuity.³⁰

DES, FED, and KCN are diseases with significant burden on patients' quality of life and on the economy. The use of deep learning autonomous diagnostic algorithms has been proposed as a solution to reduce the disease burden and to bridge the gap between health care needs and resources. Specially, autonomous diagnostic algorithms can allow the nonspecialist to detect disease with a sensitivity approximating that of a specialist, an especially important consideration in underserved areas.³¹ Given the limited number of specialized eye care providers, using such algorithms translates into patients who will be diagnosed and referred early enough for care to prevent irreversible vision loss. Our study is a proof of concept. Further work is needed to generalize our results and build a robust neural network for corneal diseases. In this study, we are reporting and validating a multidisease deep learning model that autonomously screens for

multiple common corneal diseases at once: DES, FED, and KCN. The performance of the model, using only AS-OCT images, is close to the performance of fellowship-trained American Board of Ophthalmology–certified cornea specialists who have used a battery of standard ophthalmologic examination and tests to make the clinical diagnosis. This highlights the potential utility of our algorithm in clinical practice. The multidisease network revealed excellent performance, comparable to the single-disease models, confirmed by high eye-level sensitivity values and specificity values with excellent eye-level accuracy for the diagnosis of DES, FED, KCN, and healthy categories.

In our study, we used an anterior segment OCT (AS-OCT)–based deep learning model to discriminate DES, FED, and KCN from healthy eyes. Several studies have demonstrated the utility of deep learning in the field of ophthalmology.^{32–38} Nevertheless, no studies have reported or validated multidisease models for the automated detection of the common corneal disease despite the significant disease burden.²⁰ Treder and associates evaluated a deep learning algorithm to automatically detect graft detachment after Descemet membrane endothelial keratoplasty.²⁰ Yousefi and associates,³⁹ Kovács and associates,¹⁰ Hidalgo and associates,¹¹ and Kamiya and associates¹² have used deep learning algorithms for evaluating keratoconus. Our study is novel compared with the previous studies that used deep learning and AS-OCT corneal images.^{20,39,40}

We have coupled our algorithms with AS-OCT to allow the autonomous diagnosis of corneal diseases. OCT has become a gold standard in caring for patients with retinal diseases and glaucoma. Approximately 30 million ophthalmic OCT procedures are done worldwide each year to care for these patients, which is equivalent to an OCT procedure every few seconds.^{31,41,42} The major advantages of OCT include quick, noninvasive, in vivo, quasi-histologic imaging of ocular structures. Although OCT has proven to be effective in detecting signs of corneal diseases, its use in the clinical setting has been relatively very limited as it needs highly specialized health care personnel to interpret.^{12,13,43} This highlights the utility of our deep learning algorithm in harnessing the power of OCT for the diagnosis of corneal diseases.

Our deep learning model prediction agreed with the clinical diagnosis in 125 of 132 eyes recruited in the setting of an independent clinical study. We classified the errors with respect to severity in the context of a screening examination. Of the 7 errors, 5 (3.8%) were confusion between dry eye and healthy, whereas in the sixth error a dry eye was misclassified as keratoconus. These errors would result in unnecessary referrals to an eye care provider or, in the case of 2 dry eyes classified as healthy, possible postponement of a merited referral to an eye care specialist with little likelihood of further damage to the patient's eye. Only a single error was considered a serious error, in which a keratoconus eye was misclassified as healthy. However, the other eye of the same patient was diagnosed correctly as keratoconus us-

ing our model. Thus, a referral to an eye care provider would have been made based on the patient diagnosis. The visualizations of the learned features of the model show that the model's decisions are meaningful and interpretable, which can build our trust in its performance.

Our system has very promising results. However, it is not without limitations that should be addressed in future studies. The model has been trained and tested using sufficient databases as displayed by the accuracy curves. Nevertheless, for future studies, the number of included images should be increased. Second, although some diseases can co-occur, for example, FED and DES, our work is limited to the diagnosis of the primary disease. This is likely because our OCT data for FED is not enough to train the network to discriminate between FED with DES and FED without DES. Also, use of a multilabel classification network could have introduced imbalance between the labels. In addition, there could be a clinical confusion between Fuchs dystrophy and dry eye, given the fact that both affect the epithelium and can cause similar signs and symptoms. Future study in which the physician would perform extensive dry eye testing in patients with Fuchs dystrophy to rule out coexistent dry eye is warranted to address this labeling confusion. Third, the model has not been tested using AS-OCT databases from other AS-OCT machines. Fourth, our model does not in-

corporate accessible medical data in disease detection, such as clinical history. Fifth, our model screens 3 diseases, excluding many other corneal diseases that can still affect patients. Sixth, although the algorithm is excellent at distinguishing between normal corneas and diseased corneas, the diseases were not quantified. For example, there is no differentiation between early- and late-stage FED; thus, future prospective studies are required to demonstrate the use of deep learning as a metric for measuring disease progression and help determine the therapeutic decision in different clinical scenarios. To address these limitations, we are working to build larger databases with external validation data sets, incorporate additional diseases, and develop a model that can both screen and quantify these diseases.

In conclusion, this is the first study that presents a multidisease deep learning approach to automatically discriminate common corneal diseases, namely DES, FED, and KCN, from healthy corneas using AS-OCT imaging with excellent accuracy. The reason for choosing these corneal diseases is that they are the three most common corneal diagnoses in the Bascom Palmer Eye Institute cornea clinic. We acknowledge that this is a preliminary proof of concept study of a tool that is still in development and that further studies are needed to compare this tool to clinical standards.

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All authors attest that they meet the current ICMJE criteria for authorship.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.ajo.2021.01.018](https://doi.org/10.1016/j.ajo.2021.01.018).

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